

From Baseline Repetition Codes to Detector-Aware Decoding and IBM Hardware Validation: A Three-Stage Study of Quantum-Memory Error Suppression

Project Manuscript Draft

September 21, 2026

Abstract

This paper presents a three-stage study of repetition-code quantum memory. Stage 1 establishes a Qiskit Aer baseline under bit-flip, phase-flip, and depolarizing noise, confirming that increasing code size improves logical success. Stage 2 extends the analysis to Stim circuit-level detector models with minimum-weight perfect matching (MWPM), showing that decoder choice can strongly affect logical error rates. Stage 3 validates a constrained version of the workflow on IBM Quantum hardware and evaluates short-horizon repeatability. The contribution is a layered comparison: baseline redundancy effects, decoder-structure effects, and real-device execution stability. The baseline remains the primary evidence source, while Stim and hardware are follow-on stages that answer additional questions left open by baseline-only simulations.

1 Introduction

Quantum error correction is central to scalable quantum computing because physical qubits are vulnerable to decoherence and control noise. Repetition codes provide a tractable and interpretable starting point: they encode a logical degree of freedom into multiple physical qubits and recover the logical value through redundant syndrome information.

This project is organized as a cumulative three-stage investigation rather than a single merged dataset:

1. **Stage 1 (Qiskit Aer baseline):** establish the basic scaling claim that larger code size reduces logical failure under standard noise channels.
2. **Stage 2 (Stim + MWPM):** quantify how much decoder quality changes performance in a detector-model simulation setting.
3. **Stage 3 (IBM hardware):** verify that the end-to-end workflow runs on real hardware with stable short-horizon outcomes.

2 Data Provenance and Scope

Evidence is intentionally separated by stage and stored as independent artifacts:

- Stage 1 baseline tables and figures: `qiskit-qec-noise-sim/results/*.csv`, `qiskit-qec-noise-sim/re` and `qiskit-qec-noise-sim/paper/tables/*.tex`
- Stage 2 simulation tables: `results/stim_mwpm_results.csv`, `results/decoder_comparison_results.c` and related stress-test outputs

- Stage 3 hardware tables: `results/ibm_hardware_validation_results.csv`, `results/layer3_three_run_metadata` JSON files

This separation is crucial for scientific clarity: Stage 2 and Stage 3 extend the interpretation of Stage 1; they do not replace Stage 1.

3 Methods

3.1 Stage 1: Qiskit Aer baseline

The baseline compares unprotected and repetition-encoded cases (code sizes 1, 3, 5) under bit-flip, phase-flip, and depolarizing channels. The key metric is logical success probability P_{succ} and logical error rate

$$P_{\text{err}} = 1 - P_{\text{succ}}. \quad (1)$$

This Stage 1 setup follows the original project framing: bit-flip repetition is best aligned with X-type noise, phase-flip repetition is best aligned with Z-type noise, and all results are simulation-based logical success estimates rather than hardware-level guarantees.

3.2 Stage 2: Stim with detector-model MWPM

Stim-generated repetition-memory circuits are decoded with PyMatching using detector error models. Distances $d \in \{3, 5, 7, 9, 11\}$ are evaluated across multiple physical error rates. Additional experiments probe biased noise, heterogeneous per-qubit noise, and temporal drift.

3.3 Stage 3: IBM hardware validation

A constrained protocol is executed on backend `ibm_fez` with three compact circuits (`z_idle_2`, `z_idle_4`, `x_idle_4`), 1000 shots each, and repeated jobs for stability assessment.

4 Results

4.1 Stage 1 baseline outcomes

At representative baseline operating points (bit-flip slice), logical success increases with code size ($1 \rightarrow 3 \rightarrow 5$), supporting the core redundancy claim. To retain the factual framing of the original paper, we explicitly preserve the depolarizing-channel caution: Z-basis repetition variants can be mismatched to a stored $|+\rangle$ state in that experiment configuration, yielding approximately 0.5 logical success even at $p = 0$ for those mismatched rows. This is a basis-alignment artifact, not a claim that repetition coding fails in general. For readers new to repetition-code QEC, the practical interpretation of Stage 1 is as follows. Under matched noise and basis settings, the encoded logical qubit fails less often than the unencoded baseline as physical noise increases. For example, at $p = 0.08$ in the bit-flip and phase-flip summaries, baseline $n = 1$ success is 0.7625, while repetition success is 0.8440 for $n = 3$ and 0.8915 for $n = 5$. The same monotone ordering appears at $p = 0.02, 0.04, \text{ and } 0.06$. This is the core educational message of Stage 1: larger repetition codes provide redundancy that can be converted into higher logical success, but only within the assumptions of the simulated channel model.

Table 1: Logical success probability vs physical error probability p for baseline and repetition-code error correction.

p	baseline (n=1)	repetition (n=3)	repetition (n=5)	gain n=5
0.0000	1.0000	1.0000	1.0000	0.0000
0.0200	0.9225	0.9820	0.9980	0.0755
0.0400	0.8625	0.9465	0.9755	0.1130
0.0600	0.8095	0.8900	0.9460	0.1365
0.0800	0.7625	0.8440	0.8915	0.1290

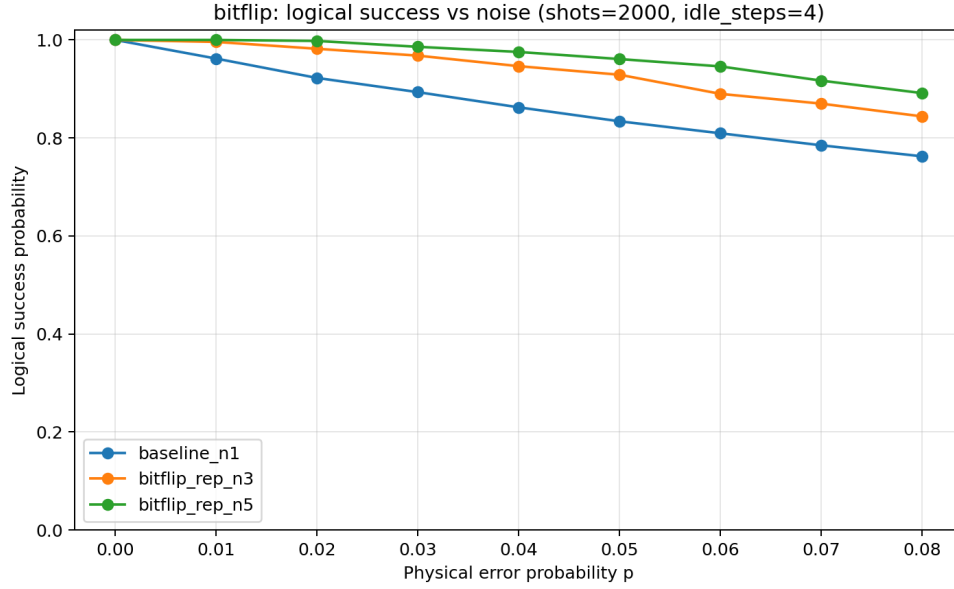


Figure 1: Stage 1 legacy baseline: bit-flip performance across code sizes (original paper asset).

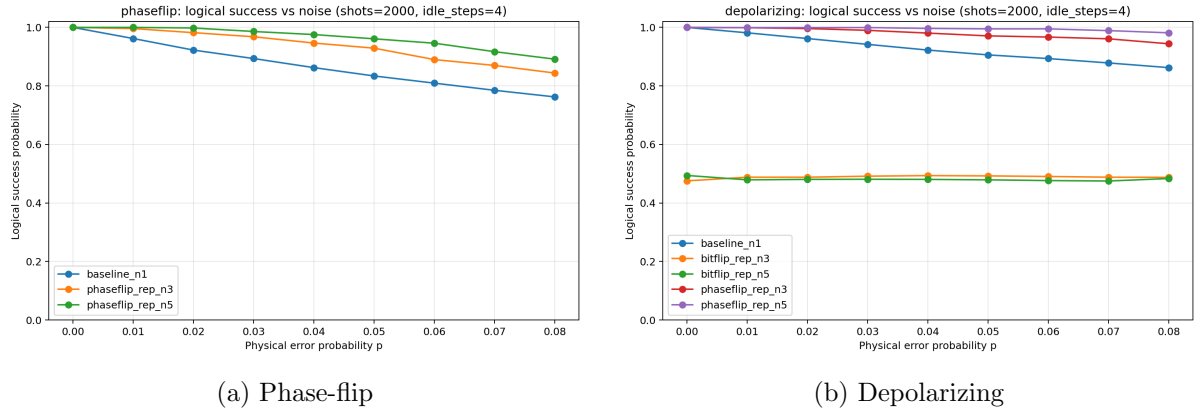


Figure 2: Stage 1 legacy baseline under additional canonical noise channels (original paper assets).

4.2 Stage 2 decoder and noise-structure outcomes

Detector-aware decoding substantially improves logical error relative to simple majority decoding in matched settings, indicating that decoding strategy is a first-order performance factor

Table 2: Logical success probability vs physical error probability p for baseline and repetition-code error correction.

p	baseline (n=1)	repetition (n=3)	repetition (n=5)	gain n=5
0.0000	1.0000	1.0000	1.0000	0.0000
0.0200	0.9225	0.9820	0.9980	0.0755
0.0400	0.8625	0.9465	0.9755	0.1130
0.0600	0.8095	0.8900	0.9460	0.1365
0.0800	0.7625	0.8440	0.8915	0.1290

Table 3: Depolarizing noise results. Note that Z-basis repetition (bitflip_rep_*) is mismatched to the stored $|+\rangle$ state in this experiment, giving ≈ 0.5 even at $p = 0$.

p	baseline_n1	phaseflip_rep_n3	phaseflip_rep_n5	bitflip_rep_n3	bitflip_rep_n5
0.0000	1.0000	1.0000	1.0000	0.4755	0.4940
0.0200	0.9620	0.9960	0.9995	0.4880	0.4805
0.0400	0.9225	0.9805	0.9965	0.4935	0.4805
0.0600	0.8935	0.9670	0.9950	0.4905	0.4765
0.0800	0.8625	0.9440	0.9815	0.4875	0.4835

rather than a minor post-processing choice. The distance-scaling curve in Fig. 3 makes the suppression trend explicit at low and moderate physical error. At $p = 0.01$, logical error decreases from 0.00681 ($d = 3$) to 0.00142 ($d = 5$), 0.00027 ($d = 7$), 0.00006 ($d = 9$), and 0.00002 ($d = 11$). At $p = 0.02$, the same ordering holds (0.0258, 0.01098, 0.00478, 0.00211, 0.00086). At higher noise ($p = 0.05$), all distances degrade and compress toward roughly 0.10–0.12 error, showing the finite operating window where code-distance gains are strongest. Decoder-comparison results in Fig. 4 quantify why Stage 2 was necessary. At $d = 11$ and $p = 0.02$, majority-vote decoding gives logical error 0.23536, while unweighted MWPM gives 0.00208 and detector-model MWPM gives 0.00112. This is more than two orders of magnitude between majority and detector-aware MWPM. Even at smaller distances, the same pattern appears (e.g., $d = 5$, $p = 0.02$: majority 0.1396 vs MWPM 0.01068). Educationally, this isolates two separable effects: encoding distance helps, and decoder quality independently helps. The bias, heterogeneity, and temporal-drift sweeps explain how non-ideal structure changes performance. In the bias sweep at $d = 3$, $p_{\text{total}} = 0.02$, error drops from 0.02402 at bias 0.1 to 0.00124 at bias 100, showing that error composition matters, not only total probability. In heterogeneous-noise stress tests, spread can either mildly shift results or sharply worsen them when effective heavy-tail behavior appears (e.g., $d = 3$, $p_{\text{mean}} = 0.05$, spread 1.0 gives 0.38847). In temporal drift snapshots, “one bad qubit” and “global drift up” snapshots consistently raise logical error relative to normal snapshots (for instance $d = 5$, $p_{\text{mean}} = 0.05$: 0.11558 normal, 0.19458 one-bad-qubit, 0.20782 global-drift-up), which motivates calibration-aware interpretations of QEC simulations.

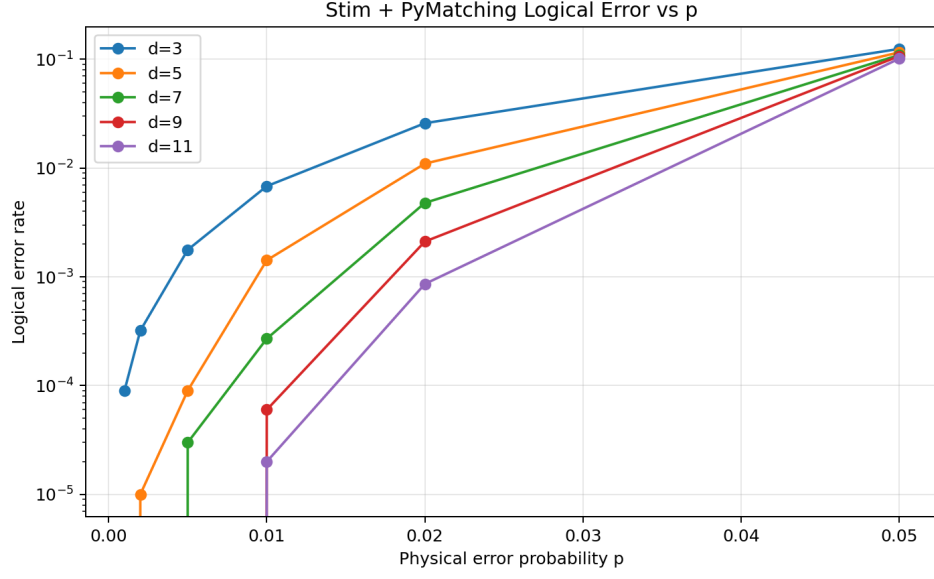


Figure 3: Stage 2 Stim MWPM distance scaling versus physical error rate.

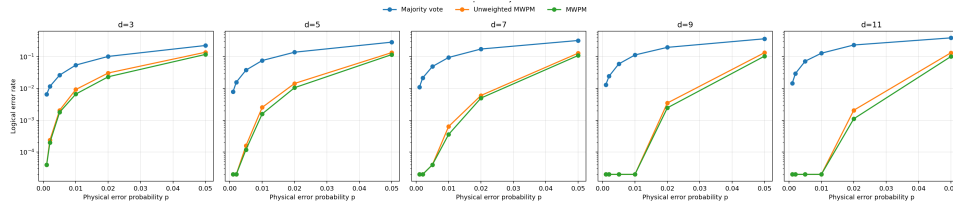


Figure 4: Stage 2 decoder comparison (majority, unweighted MWPM, detector-model MWPM).

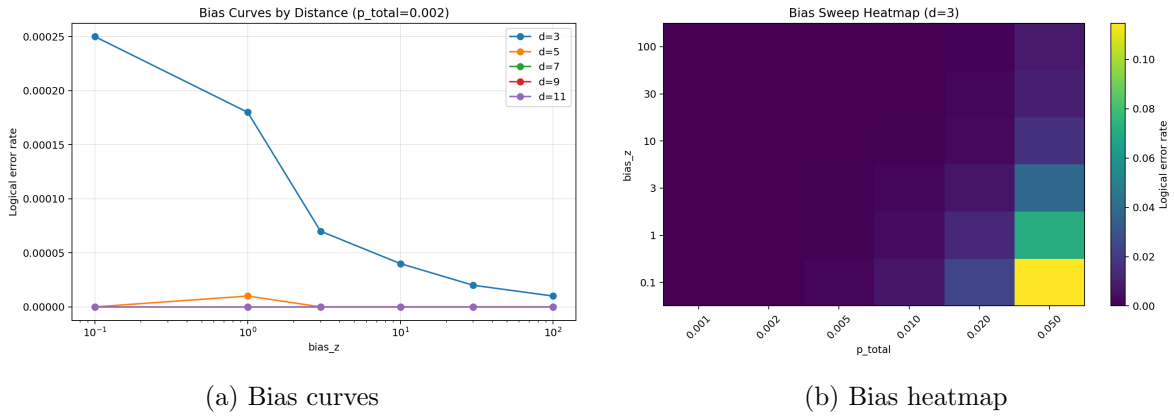


Figure 5: Stage 2 biased-noise behavior.

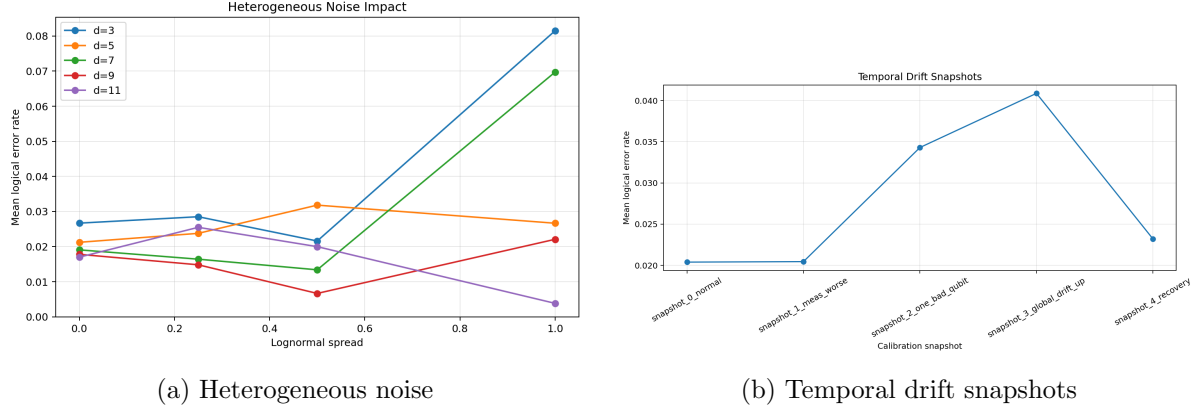


Figure 6: Stage 2 stress tests beyond i.i.d. channel assumptions.

4.3 Stage 3 hardware outcomes

Hardware validation results are near-unity logical success for the constrained circuit set, with low run-to-run variation in repeated jobs. This supports practical pipeline stability in the tested regime. Numerically, the primary hardware run reports success 0.999 for `z_idle_2` and 1.000 for `z_idle_4` and `x_idle_4`. Across three runs, the mean successes are 0.9997, 0.9997, and 1.0000 respectively, with maximum run-to-run range 0.001. This is strong evidence of short-horizon execution stability for this compact circuit family. However, this section should be read as workflow validation, not a broad claim that repetition-code QEC is solved on hardware; the circuit scope, shot count, and backend diversity are intentionally limited.

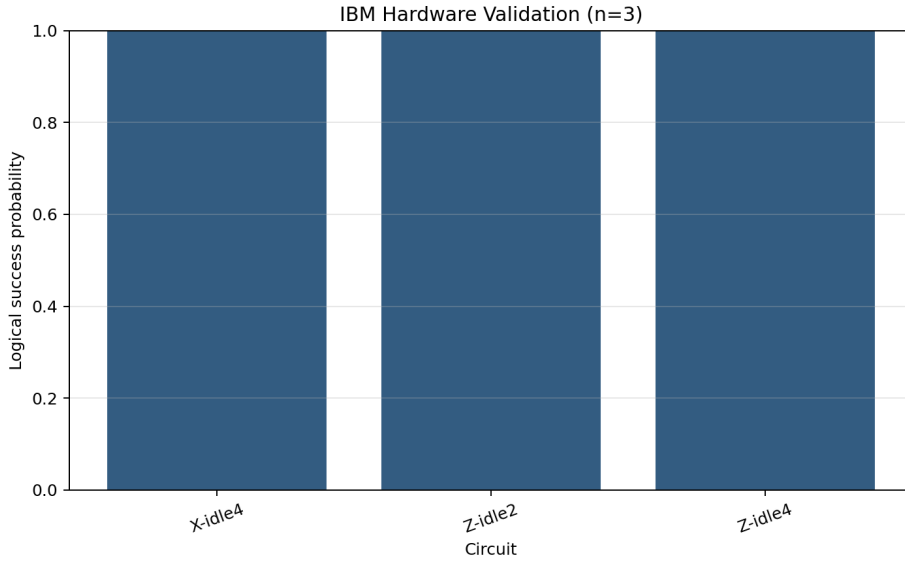


Figure 7: Stage 3 constrained IBM hardware validation outcomes.

5 Direct Cross-Stage Comparison

A single figure is used to make the pairwise comparisons explicit.

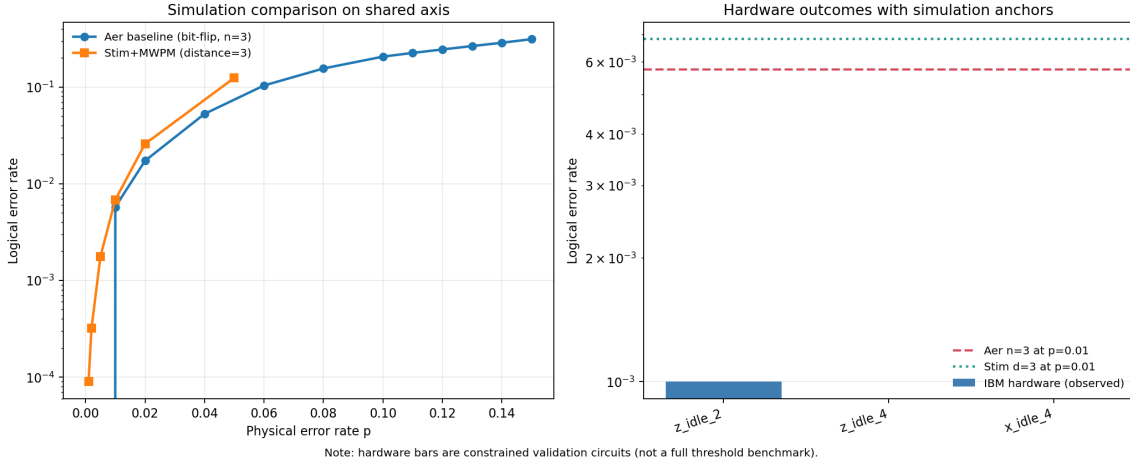


Figure 8: Direct cross-stage comparison on a shared logical-error basis. Left: Stim vs Aer (simulation-to-simulation). Right: hardware outcomes with Aer/Stim anchors for context.

5.1 Comparison basis

To avoid misleading one-to-one claims across unlike execution models, we use the following basis:

1. Shared objective: logical-memory preservation.
2. Shared outcome axis: logical error (or logical success).
3. Matched-slice simulation comparison: Aer $n = 3$ vs Stim $d = 3$.
4. Hardware interpreted as constrained validation points, not threshold-equivalent simulation replacements.

Using this basis, the pairwise comparisons can be read concretely. Aer vs Stim asks how conclusions change when moving from channel-level baseline modeling to detector-model decoding with explicit syndrome structure. At matched small-code slices, both show that lower physical noise gives lower logical error, but Stim additionally reveals large decoder-dependent separations that are hidden in simple majority-style baselines. Aer vs hardware asks whether the tested workflow remains operational outside simulation; here the answer is yes for the constrained circuits, with near-unity success and low run-to-run spread. The full three-stage narrative is therefore educationally coherent: Stage 1 teaches redundancy trends, Stage 2 teaches decoder and noise-structure effects, and Stage 3 teaches practical execution constraints.

5.2 Why Stage 2 and Stage 3 were necessary

Why Stim was necessary:

- Baseline channel sweeps established redundancy trends but did not isolate decoder-information effects.
- Detector-model MWPM quantifies the algorithmic contribution of decoder structure.

Why hardware was necessary:

- Simulation alone cannot validate cloud-runtime operational stability.
- Hardware runs test execution path, repeatability, and artifact integrity in a real backend context.

6 Discussion

The combined evidence supports three distinct conclusions:

1. Redundancy improves logical reliability (Stage 1).
2. Decoder quality can dominate performance in realistic syndrome settings (Stage 2).
3. Constrained real-device execution is stable enough to validate the workflow (Stage 3).

This layered framing is stronger than treating all outputs as a single homogeneous dataset because it attributes improvements to distinct mechanisms: encoding, decoding, and execution context. At the same time, Stage 1 should be interpreted conservatively: Aer simulations support trend-level claims under the modeled channels, but they are not by themselves sufficient evidence for broad hardware performance claims. For a repetition-code QEC audience, the key educational point is methodological: “better curve” outcomes can come from different causes, including larger code distance, better decoder inference, or easier noise composition. This is why the manuscript separates stages and reports matched comparisons wherever possible.

7 Limitations

- Repetition code protects a limited logical subspace.
- Stage 1 is a small, reproducible Qiskit/Aer simulation study; it supports comparative logical-success trends but does not establish hardware-level fault-tolerance claims.
- Some depolarizing rows in the legacy baseline intentionally document basis-mismatch behavior (approximately 0.5 success at $p = 0$ for mismatched Z-basis repetition rows); these rows are diagnostic and must not be over-interpreted as general repetition-code performance.
- Stage 3 hardware scope is intentionally narrow (single backend family, compact circuit set, limited shots).
- No threshold claim or cross-platform universality claim is made.

8 Conclusion

The project demonstrates a coherent progression from baseline simulation, to detector-aware simulation, to real-device validation. The baseline result remains foundational; Stim and hardware add necessary, non-redundant evidence. The most important technical message is that meaningful QEC evaluation requires both encoding-scale analysis and decoder-aware interpretation, with hardware used to validate operational reality.

Acknowledgments

This manuscript summarizes iterative project work on repetition-code QEC benchmarking, decoder comparison, and constrained cloud-hardware validation.

References

- [1] C. Gidney, *Stim: a fast stabilizer circuit simulator*.
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- [4] Standard references on repetition codes and majority decoding in introductory QEC literature.